

Math 161

Problem set 9 solutions

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1. If x and y are the side lengths of a rectangle, then its perimeter is given by $2x + 2y$ and its area by xy . If we fix a value $P > 0$ for the perimeter, then we have $y = \frac{1}{2}P - x$, and hence the area is given by the function

$$A(x) = x(\frac{1}{2}P - x) = \frac{1}{2}Px - x^2.$$

Certainly $x, y \geq 0$, and since

$$\frac{1}{2}P - x = y \geq 0$$

we have $x \leq \frac{1}{2}P$. Thus we are trying to find the maximum value of $A(x)$ on $[0, \frac{1}{2}P]$.

Since the function $A(x)$ is a polynomial and hence everywhere differentiable, its global maxima on $[0, \frac{1}{2}P]$ must occur at either critical points or endpoints. We have $A'(x) = \frac{1}{2}P - 2x$, so the only critical point is $x = \frac{1}{4}P$. Since $A(0) = 0 = A(\frac{1}{2}P)$, the maximum in question must occur at $x = \frac{1}{4}P$. Note that then $y = \frac{1}{4}P$ as well, so $x = y$ and this does correspond to the square of perimeter P .

2. As mentioned in the problem, we already proved existence in Problem Set 6, without any assumption on the derivative of f . We will make use of the auxiliary function $g : [0, 1] \rightarrow \mathbb{R}$ defined by $g(x) := f(x) - x$. Suppose that $x, y \in [0, 1]$ satisfy $f(x) = x$ and $f(y) = y$, and assume for the sake of contradiction that $x \neq y$. Without loss of generality, we can assume that $x < y$. Since $[x, y] \subset [0, 1]$, the function f is continuous on $[x, y]$ and differentiable on (x, y) , which implies that g also has these properties. Now the mean value theorem implies that there exists $z \in (x, y)$ such that $g'(z) = \frac{g(y) - g(x)}{y - x}$. But $g(x) = 0 = g(y)$, so we obtain

$$f'(z) - 1 = g'(z) = 0$$

and hence $f'(z) = 1$, contrary to hypothesis.

3. (a) We will assume that f is increasing. If f is decreasing, then we can simply apply the following arguments to $-f$.

First, we claim that

$$\lim_{x \rightarrow a^+} f(x) = \inf \{f(x) \mid x > a\}.$$

Note that the infimum on the right side exists: the set $\{f(x) \mid x > a\}$ is obviously nonempty, and it is bounded below by $f(a)$. Let

$$L := \inf \{f(x) \mid x > a\},$$

so we must show that for any $\epsilon > 0$, there exists $\delta > 0$ such that $a < x < a + \delta$ implies that $|f(x) - L| < \epsilon$. Since $L + \epsilon > L$, there exists $x_0 > a$ such that $f(x_0) < L + \epsilon$: otherwise $L + \epsilon$ would be a lower bound for $\{f(x) \mid x > a\}$, contradicting the fact that L is the greatest lower bound. Put $\delta := x_0 - a$, so that if $a < x < a + \delta$, we have $x < a + \delta = x_0$ and hence $f(x) \leq f(x_0) < L + \epsilon$. On the other hand, the inequality $x > a$ implies that $f(x) \geq L$ by the definition of L , so we obtain

$$L \leq f(x) < L + \epsilon,$$

which implies that $|f(x) - L| < \epsilon$ as desired.

Similarly, we have

$$\lim_{x \rightarrow a^-} f(x) = \sup \{f(x) \mid x < a\}.$$

To prove this, one can either repeat the above arguments *mutatis mutandis*, or apply them literally to $-f(2a - x)$. Namely, the latter function is increasing, and one can use the observations that

$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(2a - x) = - \lim_{x \rightarrow a^+} -f(2a - x)$$

and

$$\sup \{f(x) \mid x < a\} = - \inf \{-f(2a - x) \mid x > a\}.$$

Finally, it remains to show that

$$\lim_{x \rightarrow a^+} f(x) \neq \lim_{x \rightarrow a^-} f(x).$$

Assuming these are equal for the sake of contradiction, it follows that the corresponding two-sided limit exists and is equal to both one-sided limits. As mentioned above, the hypothesis that f is increasing implies that $f(a)$ is a lower bound for $\{f(x) \mid x > a\}$, and hence

$$f(a) \leq \inf \{f(x) \mid x > a\} = \lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a} f(x).$$

Similarly, we see that $f(a)$ is an upper bound for $\{f(x) \mid x < a\}$, which implies that

$$f(a) \geq \sup \{f(x) \mid x < a\} = \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a} f(x).$$

Combining these inequalities, we obtain that $f(a) = \lim_{x \rightarrow a} f(x)$, i.e. f is continuous at a . But this contradicts our hypothesis that f is discontinuous at a .

- (b) We continue to assume without loss of generality that f is increasing. As in the hint, let

$$A := \{a \in \mathbb{R} \mid f \text{ is discontinuous at } a\}.$$

By part (a), if $a \in A$ then f has a jump discontinuity at a . Moreover, we saw in part (a) that

$$f(a) \leq \inf \{f(x) \mid x > a\} = \lim_{x \rightarrow a^+} f(x)$$

and

$$f(a) \geq \sup \{f(x) \mid x < a\} = \lim_{x \rightarrow a^-} f(x),$$

which implies that

$$\lim_{x \rightarrow a^-} f(x) \leq f(a) \leq \lim_{x \rightarrow a^+} f(x)$$

and hence

$$\lim_{x \rightarrow a^-} f(x) < \lim_{x \rightarrow a^+} f(x).$$

By the density of $\mathbb{Q}$ in $\mathbb{R}$, for any $a \in A$ we can therefore choose $g(a) \in \mathbb{Q}$ such that

$$\lim_{x \rightarrow a^-} f(x) < g(a) < \lim_{x \rightarrow a^+} f(x).$$

This defines a map $g : A \rightarrow \mathbb{Q}$. We claim that g is injective: then we will be done, since $\mathbb{Q}$ is countably infinite and therefore any subset of $\mathbb{Q}$ is either finite or countably infinite. Choose $a, b \in A$ such that $a \neq b$, so we must show that $g(a) \neq g(b)$ (if A does not contain two distinct elements then it is a singleton or empty, in which case g is automatically injective). Without loss of generality, we can assume that $a < b$. Choosing $x_0 \in \mathbb{R}$ such that $a < x_0 < b$, we have

$$\lim_{x \rightarrow a^+} f(x) = \inf \{f(x) \mid x > a\} \leq f(x_0) \leq \sup \{f(x) \mid x < b\} = \lim_{x \rightarrow b^-} f(x).$$

Thus we have

$$g(a) < \lim_{x \rightarrow a^+} f(x) \leq \lim_{x \rightarrow b^-} f(x) < g(b),$$

so $g(a) \neq g(b)$ as desired.

4. (a) Consider the function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = -x^5 - x$. We have $g'(x) = -5x^4 - 1 < 0$ for all $x \in \mathbb{R}$, so the function g is strictly decreasing and hence injective. It is also surjective: for any $a \in \mathbb{R}$, the equation $g(x) = -x^5 - x = a$ is equivalent to $x^5 + x - a = 0$, and we showed on Problem Set 6 that any polynomial of odd degree has a root.

Thus g admits an inverse $g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$. Since $g'(x) \neq 0$ for all $x \in \mathbb{R}$, the inverse function theorem implies that $f := g^{-1}$ is differentiable everywhere. Moreover, we have

$$f(x)^5 + f(x) + x = -g(f(x)) + x = -x + x = 0$$

as needed.

- (b) There are two ways to do this: one is to differentiate the equation $f(x)^5 + f(x) + x = 0$ to obtain

$$5f(x)^4 f'(x) + f'(x) + 1 = \frac{d}{dx} \left(f(x)^5 + f(x) + x \right) = 0$$

and hence

$$f'(x) = \frac{-1}{5f(x)^4 + 1}$$

(note that the denominator $5f(x)^4 + 1 > 0$ for all $x \in \mathbb{R}$, so this formula makes sense). Alternatively, one can apply the inverse function theorem to find that

$$f'(x) = f'(g(f(x))) = \frac{1}{g'(f(x))} = \frac{1}{-5f(x)^4 - 1}$$

(here $f = g^{-1}$ as in part (a)).